

GROUP ASSIGNMENT



TOPIC : STUDY OF SIGNALISED INTERSECTION BY INDO HCM METHOD

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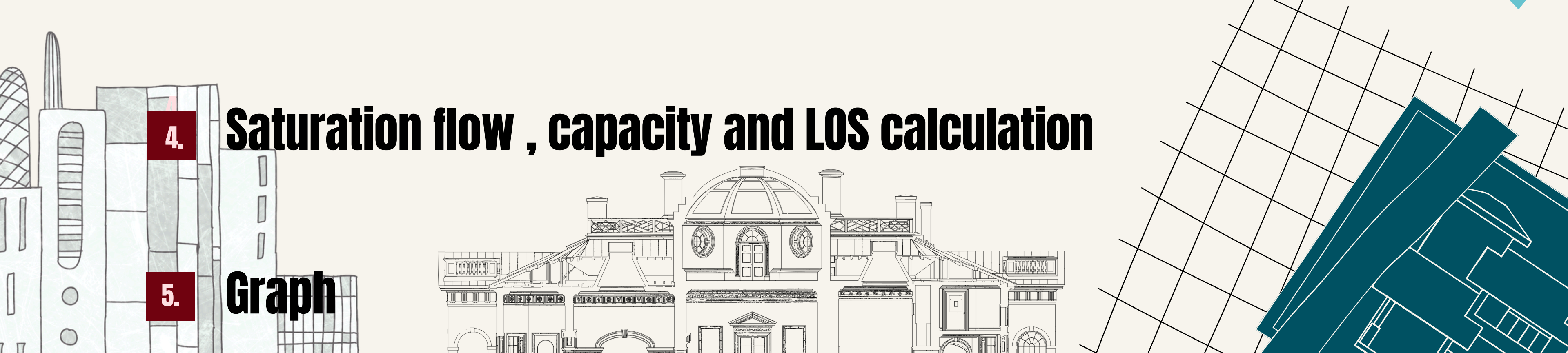
1. Introduction

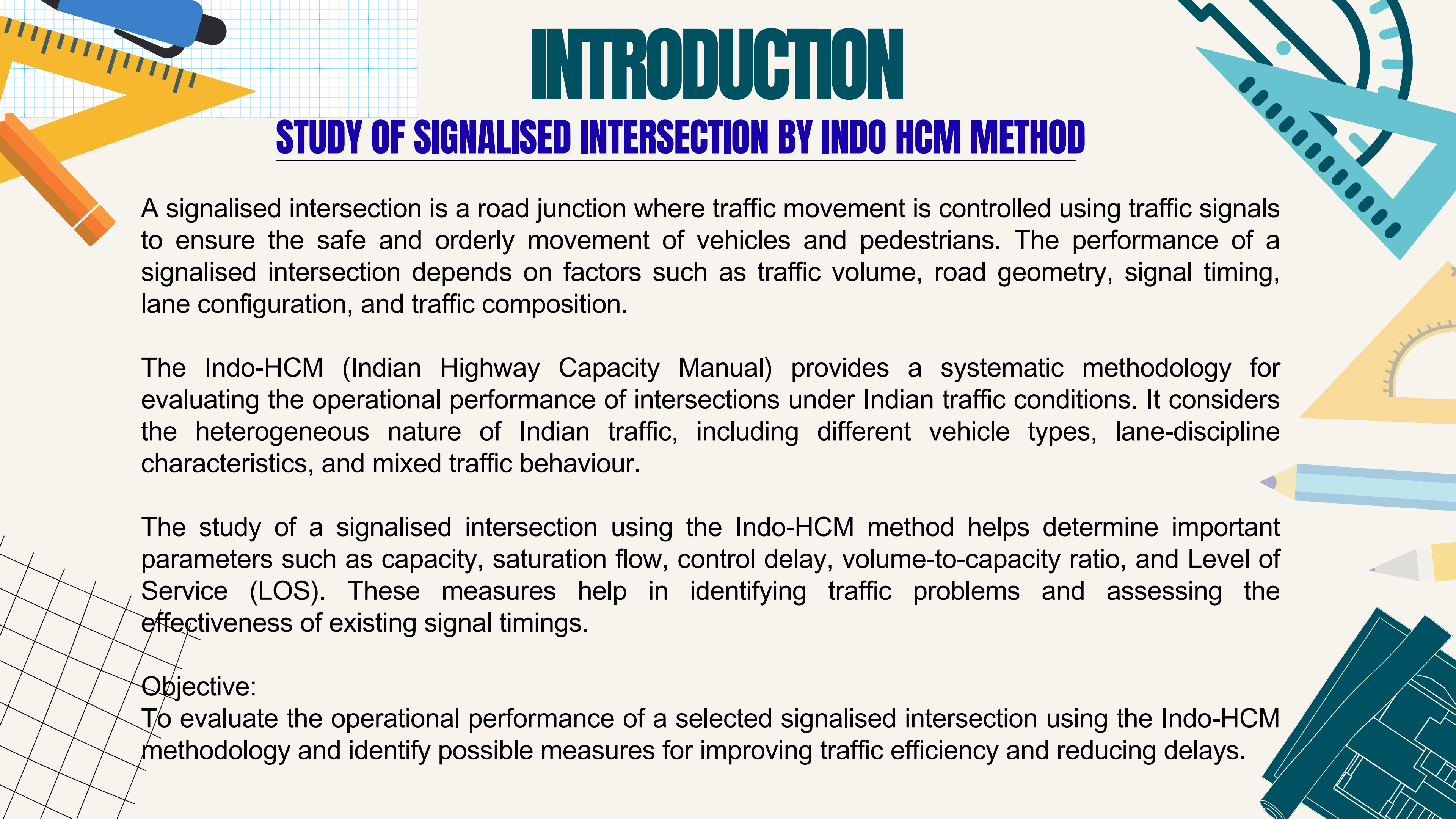
2. Location details

3. Vehicular data for each bound

4. Saturation flow , capacity and LOS calculation

5. Graph





INTRODUCTION

STUDY OF SIGNALISED INTERSECTION BY INDO HCM METHOD

A signalised intersection is a road junction where traffic movement is controlled using traffic signals to ensure the safe and orderly movement of vehicles and pedestrians. The performance of a signalised intersection depends on factors such as traffic volume, road geometry, signal timing, lane configuration, and traffic composition.

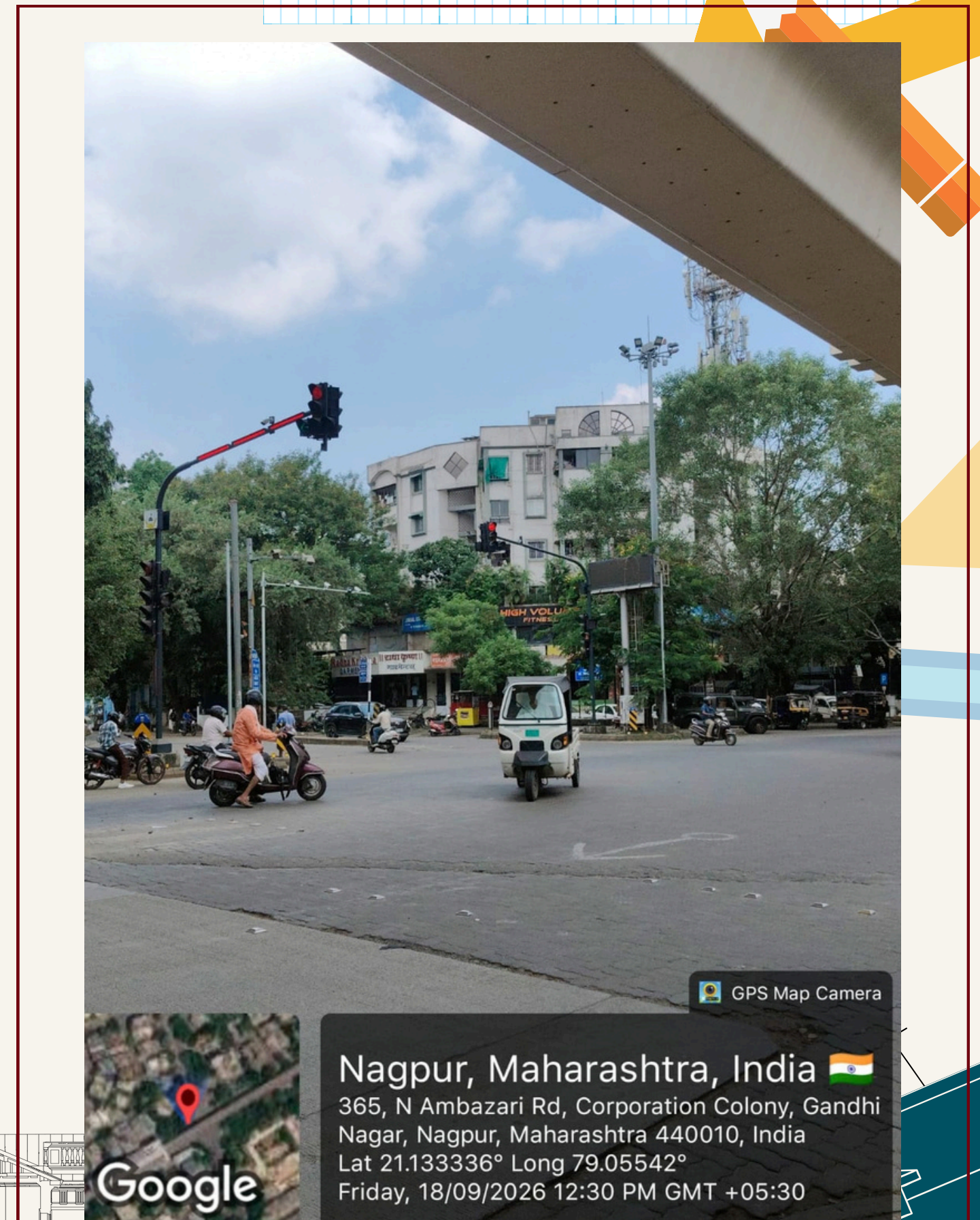
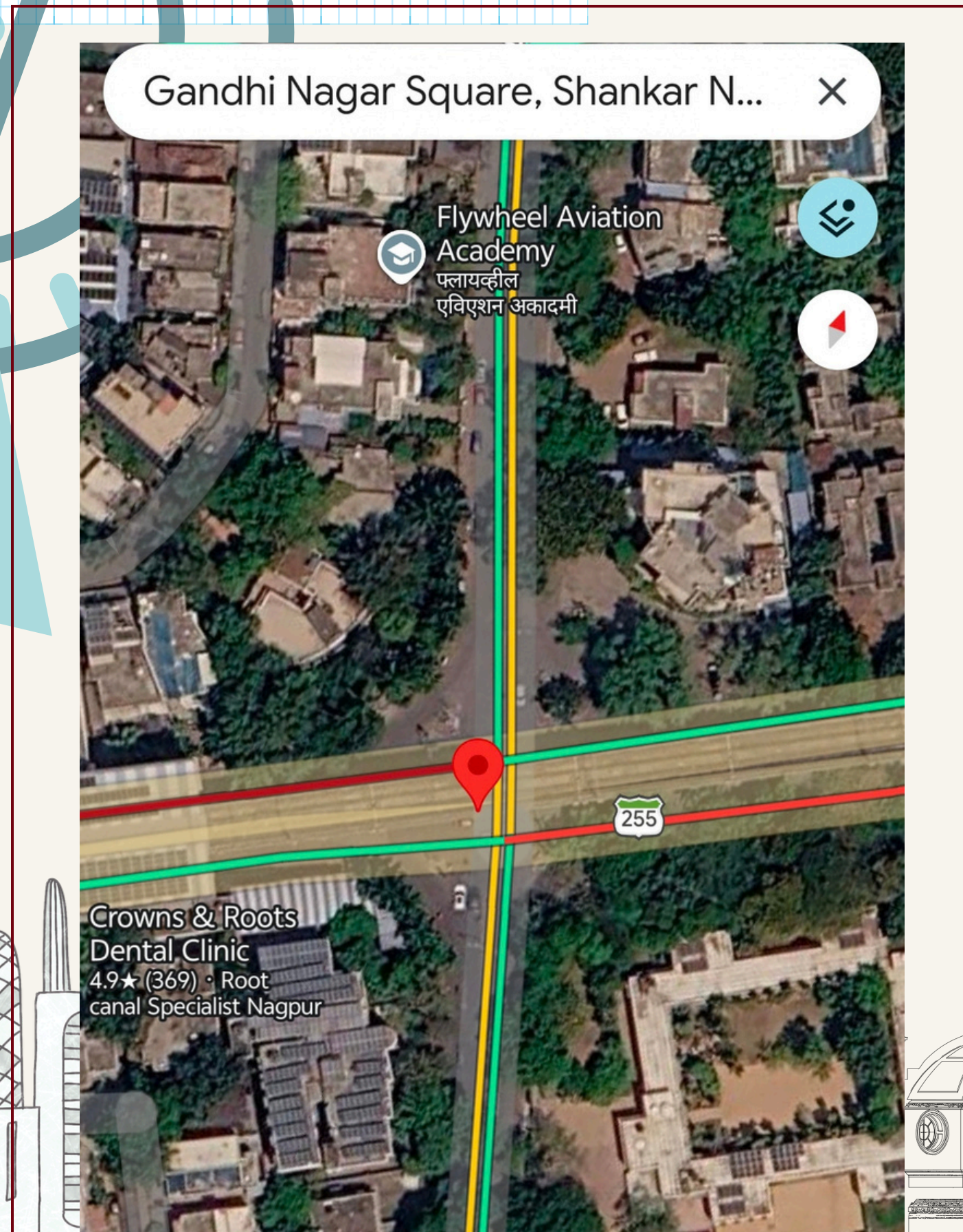
The Indo-HCM (Indian Highway Capacity Manual) provides a systematic methodology for evaluating the operational performance of intersections under Indian traffic conditions. It considers the heterogeneous nature of Indian traffic, including different vehicle types, lane-discipline characteristics, and mixed traffic behaviour.

The study of a signalised intersection using the Indo-HCM method helps determine important parameters such as capacity, saturation flow, control delay, volume-to-capacity ratio, and Level of Service (LOS). These measures help in identifying traffic problems and assessing the effectiveness of existing signal timings.

Objective:

To evaluate the operational performance of a selected signalised intersection using the Indo-HCM methodology and identify possible measures for improving traffic efficiency and reducing delays.

LOCATION : GANDHI NAGAR SQUARE (LAD SQUARE)



TEAM



ESTIMATION OF SATURATION FLOW

$$SF = w \times USF_0 \times f_{bb} \times f_{br} \times f_{is}$$

$$USF_0 = \begin{cases} 630; & \text{for } w < 7.0m \\ 1140 - 60w; & \text{for } 7.0 \leq w \leq 10.5m \\ 500; & \text{for } w > 10.5m \end{cases}$$

Where,

USF₀ = Unit base saturation flow rate (in PCU / hour / m)
w = effective width of approach in meters (m).

Adjustment factor for bus blockage Adjustment factor for right turning vehicles

$$f_{bb} = \frac{w - 3\left(\frac{t_b n_B}{3600}\right)}{w} \qquad f_{br} = \frac{w - w_r}{w}$$

Adjustment factor for initial surge of vehicles

Green Time of Phase (seconds)	Only Anticipation Effect	Only Approach Flare Effect		Anticipation and Approach Flare Effect	
		Low (S _R =1.15)	High (S _R =1.35)	Low (S _R =1.15)	High (S _R =1.35)
< 15	1.133	1.020	1.047	1.153	1.180
15 - 30	1.067	1.010	1.023	1.077	1.090
30 - 45	1.044	1.007	1.016	1.051	1.060
45 - 60	1.033	1.005	1.012	1.038	1.045
60 - 75	1.027	1.004	1.009	1.031	1.036
75 - 90	1.022	1.003	1.008	1.026	1.030
90 - 120	1.017	1.002	1.006	1.019	1.023
> 120	1.000	1.000	1.000	1.000	1.000

NORTH BOUND

NORTH BOUND												
Cycle	Time Stamp	Type of Vehicles	No.of Vehicles	PCU Factor	PCU		Cycle	Time Stamp	Type of Vehi	No.of Vehicle	PCU Factor	PCU
C-1	00:05-00:35	2-Wheeler	24	0.4	9.6		C-2	02:05 – 02:3	2-Wheeler	21	0.4	8.4
		3-Wheeler	7	0.5	3.5				3-Wheeler	6	0.5	3
		Standard Car	10	1	10				Standard Car	9	1	9
		Buses	1	1.6	1.6				Buses	0	1.6	0
		LCV	2	1.1	2.2				LCV	1	1.1	1.1
		HCV	1	1.6	1.6				HCV	0	1.6	0
		Bicycle	2	0.3	0.6				Bicycle	1	0.3	0.3
		Bullock Carts	0	4	0				Bullock Carts	0	4	0
		Total Vehicles	47		29.1				Total Vehicles	38		21.8
Demand Volume			873		Demand Volume			654				
Cycle	Time Stamp	Type of Vehicles	No.of Vehicles	PCU Factor	PCU		Cycle	Time Stamp	Type of Vehi	No.of Vehicle	PCU Factor	PCU
C-3	04:05 – 04:3	2-Wheeler	26	0.4	10.4		C-4	06:05 – 06:3	2-Wheeler	23	0.4	9.2
		3-Wheeler	8	0.5	4				3-Wheeler	5	0.5	2.5
		Standard Car	12	1	12				Standard Car	11	1	11
		Buses	1	1.6	1.6				Buses	1	1.6	1.6
		LCV	2	1.1	2.2				LCV	2	1.1	2.2
		HCV	1	1.6	1.6				HCV	0	1.6	0
		Bicycle	2	0.3	0.6				Bicycle	1	0.3	0.3
		Bullock Carts	0	4	0				Bullock Carts	0	4	0
		Total Vehicles	52		32.4				Total Vehicles	46		26.8
Demand Volume			972		Demand Volume			804				
Cycle	Time Stamp	Type of Vehicles	No.of Vehicles	PCU Factor	PCU							
C-5	08:05 – 08:3	2-Wheeler	25	0.4	10							
		3-Wheeler	7	0.5	3.5							
		Standard Car	9	1	9							
		Buses	0	1.6	0							
		LCV	4	1.1	4.4							
		HCV	1	1.6	1.6							
		Bicycle	1	0.3	0.3							
		Bullock Carts	0	4	0							
		Total Vehicles	47		28.8							
Demand Volume			864									
Avg Demand Volume								797.4PCU/hr				

[illegible]

SOUTH BOUND													
Cycle	Time Stamp	Type of Vehicles	No.of Vehicles	PCU Factor	PCU		Cycle	Time Stamp	Type of Vehicle	No.of Vehicle	PCU Factor	PCU	
C-1	00:05-00:35	2-Wheeler	28	0.4	11.2		C-2	02:05 – 02:35	2-Wheeler	25	0.4	10	
		3-Wheeler	5	0.5	2.5				3-Wheeler	7	0.5	3.5	
		Standard Car	9	1	9				Standard Car	12	1	12	
		Buses	0	1.6	0				Buses	1	1.6	1.6	
		LCV	3	1.1	3.3				LCV	1	1.1	1.1	
		HCV	1	1.6	1.6				HCV	1	1.6	1.6	
		Bicycle	2	0.3	0.6				Bicycle	1	0.3	0.3	
		Bullock Carts	0	4	0				Bullock Carts	0	4	0	
		Total Vehicles	48		28.2				Total Vehicles	48		30.1	
		Demand Volume			846				Demand Volume			903	
Cycle	Time Stamp	Type of Vehicles	No.of Vehicles	PCU Factor	PCU		Cycle	Time Stamp	Type of Vehicle	No.of Vehicle	PCU Factor	PCU	
C-3	04:05 – 04:35	2-Wheeler	30	0.4	12		C-4	06:05 – 06:35	2-Wheeler	27	0.4	10.8	
		3-Wheeler	6	0.5	3				3-Wheeler	8	0.5	4	
		Standard Car	10	1	10				Standard Car	11	1	11	
		Buses	1	1.6	1.6				Buses	0	1.6	0	
		LCV	5	1.1	5.5				LCV	4	1.1	4.4	
		HCV	0	1.6	0				HCV	1	1.6	1.6	
		Bicycle	2	0.3	0.6				Bicycle	1	0.3	0.3	
		Bullock Carts	0	4	0				Bullock Carts	0	4	0	
		Total Vehicles	54		32.7				Total Vehicles	52		32.1	
		Demand Volume			981				Demand Volume			963	
Cycle	Time Stamp	Type of Vehicles	No.of Vehicles	PCU Factor	PCU								
C-5	08:05 – 08:35	2-Wheeler	22	0.4	8.8								
		3-Wheeler	8	0.5	4								
		Standard Car	7	1	7								
		Buses	1	1.6	1.6								
		LCV	2	1.1	2.2								
		HCV	1	1.6	1.6								
		Bicycle	1	0.3	0.3								
		Bullock Carts	0	4	0								
		Total Vehicles	42		25.5		Avg Demand Volume			801.6			
		Demand Volume			765								

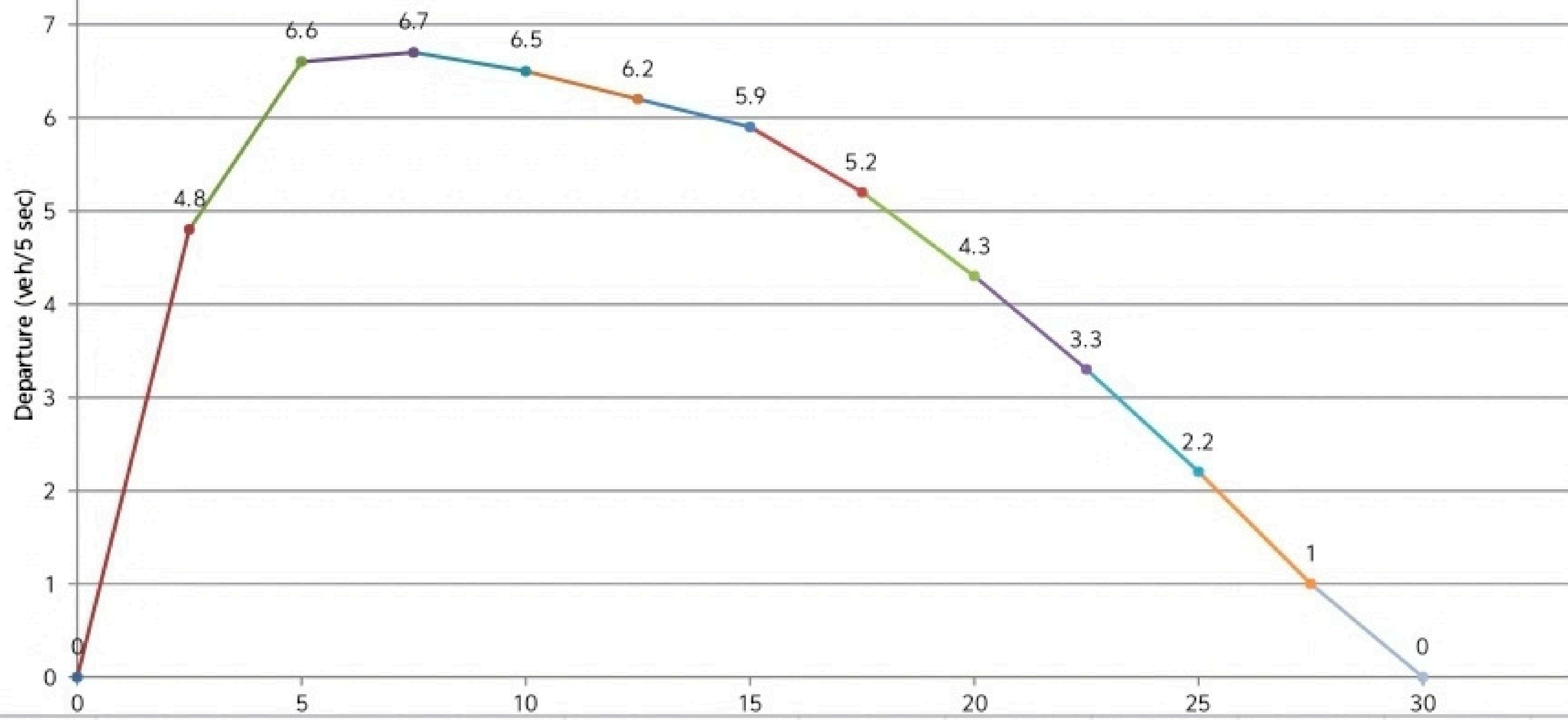
		WEST BOUND											
	Cycle	Time Stamp	Type of Vehicles	No.of Vehicles	PCU Factor	PCU		Cycle	Time Stamp	Type of Vehicle	No.of Vehicle	PCU Factor	PCU
	C-1	00:05-00:35	2-Wheeler	16	0.4	6.4		C-2	02:05 – 02:35	2-Wheeler	15	0.4	6
			3-Wheeler	8	0.5	4				3-Wheeler	10	0.5	5
			Standard Car	14	1	14				Standard Car	9	1	9
			Buses	0	1.6	0				Buses	0	1.6	0
			LCV	1	1.1	1.1				LCV	4	1.1	4.4
			HCV	1	1.6	1.6				HCV	0	1.6	0
			Bicycle	1	0.3	0.3				Bicycle	2	0.3	0.6
			Bullock Carts	0	4	0				Bullock Carts	0	4	0
			Total Vehicles	41		27.4				Total Vehicles	40		25
			Demand Volume			822				Demand Volume			750
	Cycle	Time Stamp	Type of Vehicles	No.of Vehicles	PCU Factor	PCU		Cycle	Time Stamp	Type of Vehicle	No.of Vehicle	PCU Factor	PCU
	C-3	04:05 – 04:35	2-Wheeler	17	0.4	6.8		C-4	06:05 – 06:35	2-Wheeler	12	0.4	4.8
			3-Wheeler	7	0.5	3.5				3-Wheeler	9	0.5	4.5
			Standard Car	9	1	9				Standard Car	9	1	9
			Buses	1	1.6	1.6				Buses	0	1.6	0
			LCV	3	1.1	3.3				LCV	5	1.1	5.5
			HCV	1	1.6	1.6				HCV	0	1.6	0
			Bicycle	1	0.3	0.3				Bicycle	1	0.3	0.3
			Bullock Carts	0	4	0				Bullock Carts	0	4	0
			Total Vehicles	39		26.1				Total Vehicles	36		24.1
			Demand Volume			783				Demand Volume			723
	Cycle	Time Stamp	Type of Vehicles	No.of Vehicles	PCU Factor	PCU							
	C-5	08:05 – 08:35	2-Wheeler	15	0.4	6							
			3-Wheeler	8	0.5	4							
			Standard Car	11	1	11							
			Buses	0	1.6	0							
			LCV	4	1.1	4.4	Avg Demand Volume			781.2			
			HCV	1	1.6	1.6							
			Bicycle	2	0.3	0.6							
			Bullock Carts	0	4	0							
			Total Vehicles	41		27.6							
			Demand Volume			828							

157							
158			NORTH	EAST	SOUTH	WEST	
159		PCU/hr	797.4	733.94	801.6	781.2	
160		Lane width	10.5	7	10.5	7	
161		USF.	510	720	510	720	
162		Fbb	1	1	1	1	
163		Fbr	0.64	0.64	0.64	0.64	
164		Fis	1	1	1	1	
165		Saturation flow	3427.2	3225.6	3427.2	3225.6	
166		Capacity (C)	856.8	806.4	856.8	806.4	
167							
168		X =V/C	0.930672268	0.9101438492	0.9355742297	0.96875	
169		D1	43.97	44.7	44.7	59.4	
170		D2	5.88	7.59	7.59	8.82	
171		total dealy D	45.45	47.82	47.82	62.28	
172							
173							
174				AVERAGE DELA	50.81		
175							
176				LOS= C			
177							
178							

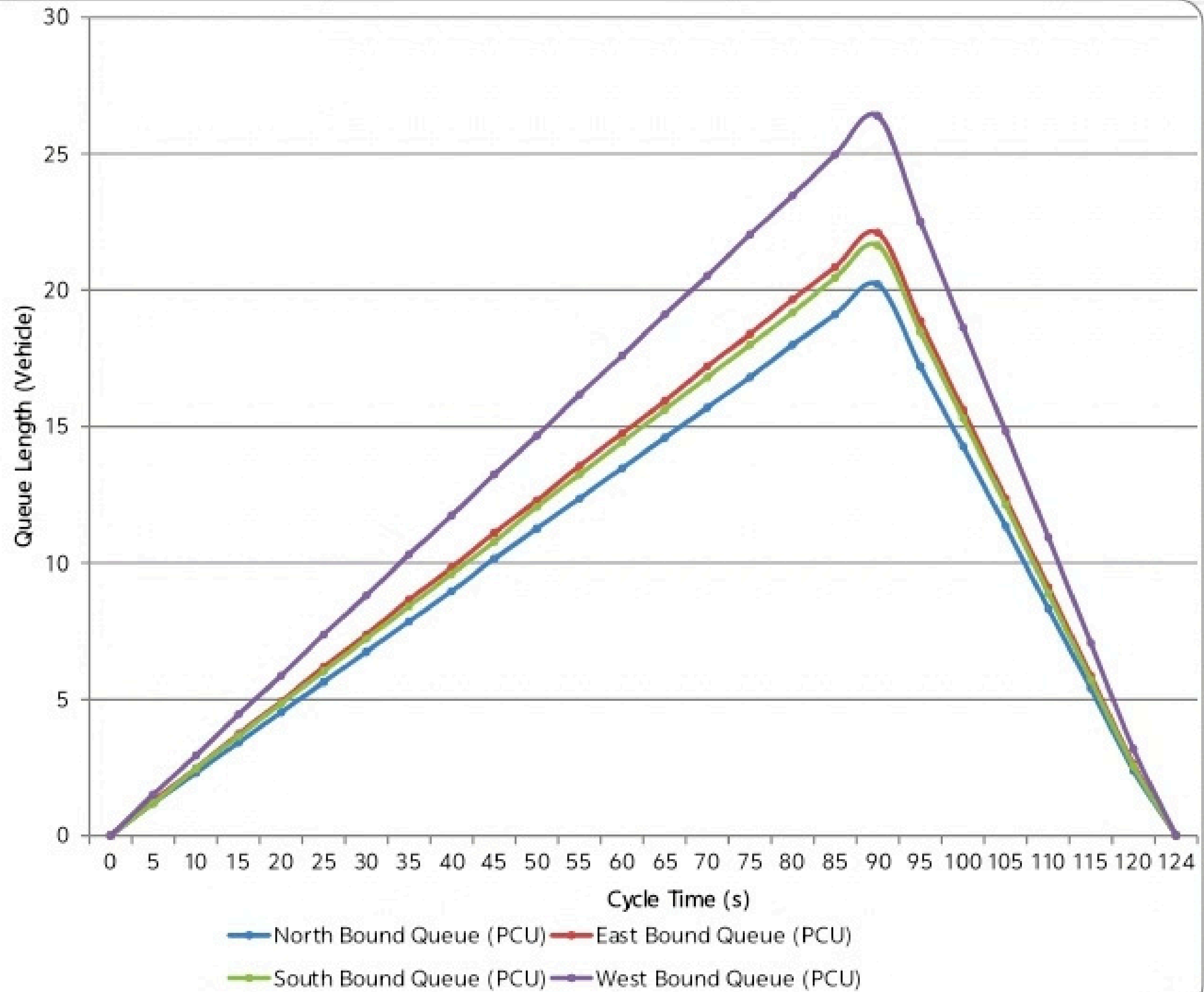
Indo-HCM Pr...

A	B	C	D	E
INDO-HCM FLOW PROFILE				
<i>Illustrative departure profile for a 30 s green signal</i>				
Green Time (s)	Departure (veh/5 sec)		Parameter	Value
0	0		Actual green	30
2.5	4.8		Illustrative effective green start	2.5
5	6.6		Illustrative effective green end	27.5
7.5	6.7		Effective green, q_e	25
10	6.5			
12.5	6.2			
15	5.9		Note Illustrative data only Shape follows the Indo-HCM departure-flow profile.	
17.5	5.2			
20	4.3			
22.5	3.3			
25	2.2			
27.5	1			
30	0			

Departure vs Effective Green Time



1	O-HCM STYLE QUEUE PROFILE					
2	s Green = 30 s Amber = 4 s Cycle = 124 s					
3	Red =90 s					
4	Cycle Time (s)	North Bound Queue (PCU)	East Bound Queue (PCU)	South Bound Queue (PCU)	West Bound Queue (PCU)	
5	0	0	0	0	0	
6	5	1.12	1.226	1.198	1.466	
7	10	2.24	2.452	2.397	2.932	
8	15	3.36	3.677	3.595	4.398	
9	20	4.481	4.903	4.794	5.865	
10	25	5.601	6.129	5.992	7.331	
11	30	6.721	7.355	7.19	8.797	
12	35	7.841	8.581	8.389	10.263	
13	40	8.961	9.806	9.587	11.729	
14	45	10.081	11.032	10.785	13.195	
15	50	11.202	12.258	11.984	14.661	
16	55	12.322	13.484	13.182	16.127	
17	60	13.442	14.71	14.381	17.594	
18	65	14.562	15.935	15.579	19.06	
19	70	15.682	17.161	16.777	20.526	
20	75	16.802	18.387	17.976	21.992	
21	80	17.923	19.613	19.174	23.458	
22	85	19.043	20.839	20.373	24.924	
23	90	20.163	22.065	21.571	26.39	
24	95	17.198	18.82	18.399	22.509	
25	100	14.233	15.575	15.227	18.628	
26	105	11.268	12.33	12.054	14.748	
27	110	8.302	9.085	8.882	10.867	
28	115	5.337	5.841	5.71	6.986	
29	120	2.372	2.596	2.538	3.105	
30	124	0	0	0	0	
31						
32						
33						
34	Approach	Avg PCU/cycle	Arrival rate (PCU/s)	Queue at start of green (PCU)	Discharge rate (PCU/s)	Queue clears at (s)
35	North Bound	27.78	0.224	20.163	0.8171	124
36	East Bound	30.4	0.2452	22.065	0.8941	124
37	South Bound	29.72	0.2397	21.571	0.8741	124
38	West Bound	36.36	0.2932	26.39	1.0694	124
39						
40						





THANK YOU